

CSCI 432 Handout 12: Shortest Paths in Linearly Interpolated Graphs

Name: class

2 April 2026

Problem 1 (SP in Linearly Interpolated Graphs). Given two weighted, directed graphs $G_0 = (V, E, \omega_0)$ and $G_1 = (V, E, \omega_1)$, define the linear interpolation at parameter $\tau \in [0, 1]$ to be

$$G_\tau := (V, E, \omega_\tau := \tau\omega_1 + (1 - \tau)\omega_0).$$

Given a source $s \in V$ and a sink $t \in V$, we want a data structure that will efficiently answer the query of: for $\tau \in [0, 1]$, what is the shortest path from s to t ?

Recall Dijkstra's algorithm from our last class period. Here, we provide pseudocode for Dijkstra's SP algorithm for easy reference (this computes just the distance; can you adjust this to find the path too?):¹

Algorithm 1 Dijkstra's Shortest-Path Algorithm

Input: A weighted, directed graph $G = (V, E, \omega)$ with non-negative edge weights; a source vertex $s \in V$

Output: Shortest-path distances $\text{dist}(v)$ for all $v \in V$

- 1: Initialize dist , a data structure indexed by V , with value ∞ for all vertices except $\text{dist}(s) = 0$
 - 2: Q min-priority queue storing the elements of V , with $\text{dist}[v]$ the priority of v for all v
 - 3: **while** $Q \neq \emptyset$ **do**
 - 4: $u \leftarrow Q.\text{pop}()$
 - 5: **for each** (u, v) out of u **do**
 - 6: $\text{alt} \leftarrow \text{dist}(u) + \omega(u, v)$
 - 7: **if** $\text{alt} < \text{dist}(v)$ **then**
 - 8: $\text{dist}[v] \leftarrow \text{alt}$
 - 9: $Q.\text{DECREASE-KEY}(v, \text{dist}[v])$
 - 10: **end if**
 - 11: **end for**
 - 12: **end while**
 - 13: **return** dist
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¹Initial pseudocode was generated from CatChat, using the prompt "simple pseudocode for dijkstra in latex using the algorithmic package (not algpseudocode)." The output was far from perfect, so what you see here is an edited version of that output.

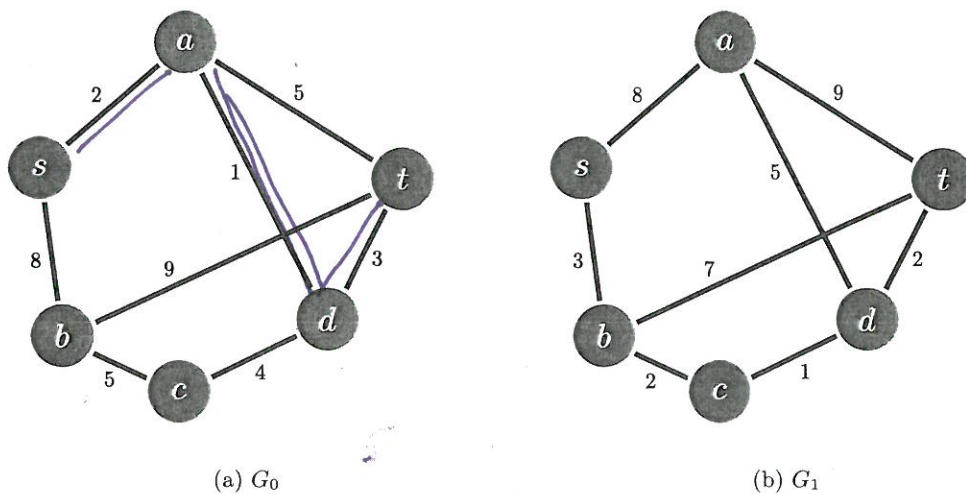


Figure 1: The starting and ending graphs that we linearly interpolate between. Here, let all edges be directed from left to right.

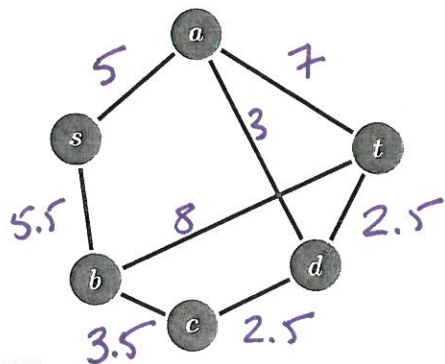
Understanding the Problem

1. For G_0 and G_1 , use Dijkstra's algorithm to compute a shortest path from s to t in each graph (denote them by p_0 and p_1 , respectively). What are their lengths (denote them by ℓ_0 and ℓ_1 , respectively).

Answer

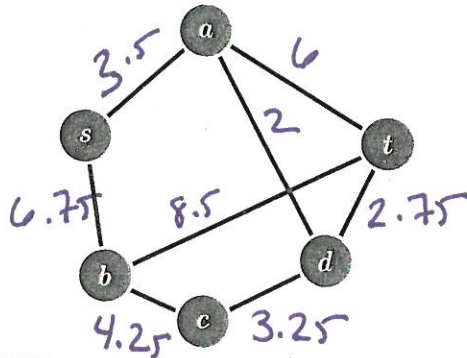
$$\begin{aligned}
 p_0 &= s a d t \\
 \ell_0 &= c_0(p_0) = 6 \\
 c_1(p_0) &= 15
 \end{aligned}$$

2. Compute $G_{0.5}$ and find a shortest path from s to t . What is its length? (Denote the path and length by $p_{0.5}$ and $\ell_{0.5}$).



Answer

3. Compute $G_{0.25}$ and find the length of a shortest path from s to t . What is its length?



Hint: Half way bwn $G_{0.5}$ and G_0

Answer

4. For which value of τ are the lengths of p_0 and p_1 the same (Note: there is one value τ for which this holds; we denote this by t_{same})?

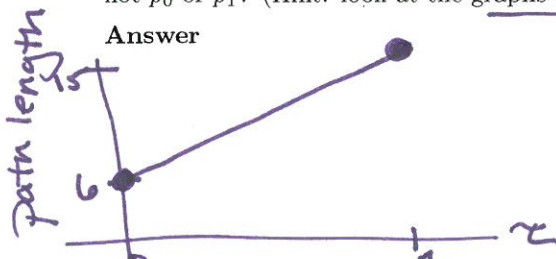
Answer

Hint: The sum of linear fns is a linear fn

$$C_\tau(p_0) = C_\tau(p_1) = \begin{cases} 6, & \tau=0 \\ \tau \cdot 15 + (1-\tau) \cdot 6, & \tau \in (0,1) \\ 15, & \tau=1 \end{cases}$$

5. What are the values of all paths between s and t at t_{same} . Is there any time that the shortest path is not p_0 or p_1 ? (Hint: look at the graphs of path length versus parameter τ for all paths from s to t).

Answer



$$C_0(p_0) = 6, C_1(p_0) = 15$$

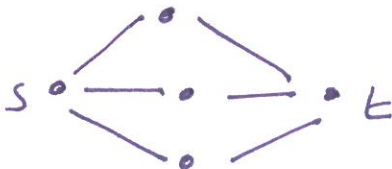
(from Q1)

At $\tau = 2/3$: Both p_0 and p_1 are shortest paths.

6. Provide an example of two graphs for which there exists a τ for which $p_t \notin \{p_0, p_1\}$.

Answer

"Easier" graph to find ^{such} weights:



$$\begin{aligned} C_\tau(p_0) &= C_\tau(p_1) \\ 15\tau + (1-\tau)6 &= \tau \cdot 8 + (1-\tau)20 \\ \vdots \\ \tau &= 2/3 \approx 0.666 \end{aligned}$$

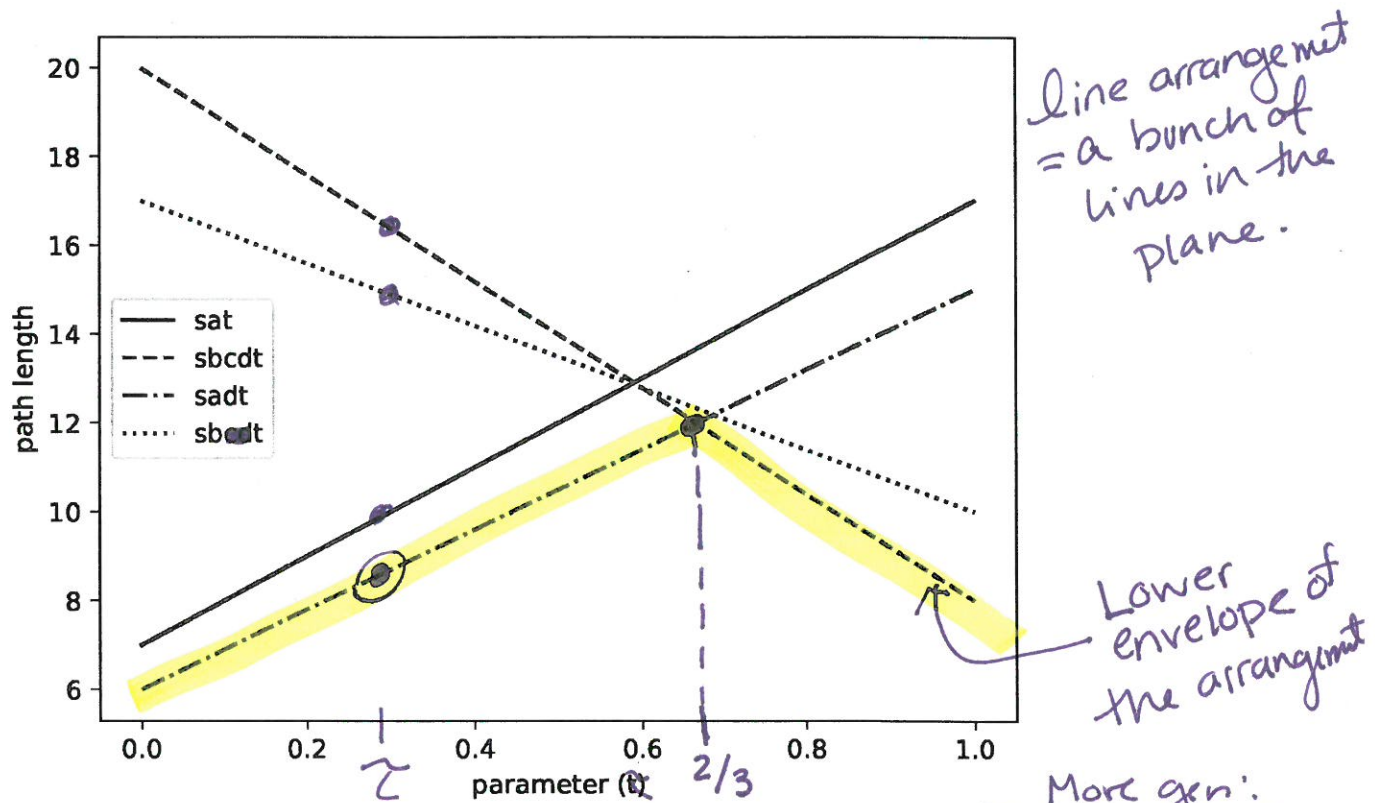


Figure 2: Path lengths.

The Algorithm

1. What are some observations we have about the plot above? When are the shortest paths not unique?

Answer

- each path's cost is linear $\Rightarrow$ the graph of it is a line (segment)

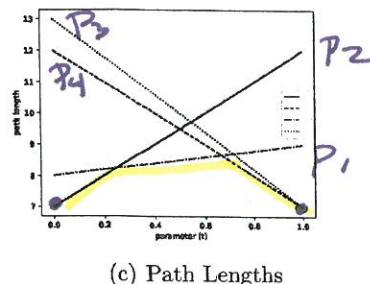
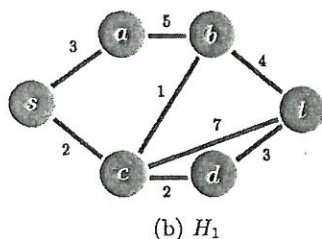
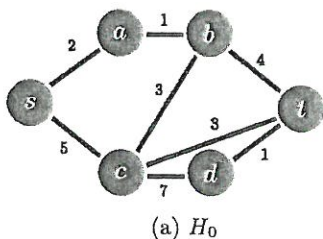


Figure 3: The starting and ending graphs that we linearly interpolate between. Here, let all edges be directed from left to right.

2. Consider the linearly interpolated graph in Figure 3, along with the graph of the lengths of all paths from s to t . What are the shortest path representatives for each parameter τ ?

Answer

Algo Questions:

- ① How do I compute the L.E. w/out explicitly listing / touching all shortest paths b/w s & t ?

~~Diagram of a graph with multiple paths~~ ← this path doesn't matter!
Why not?



- ② How do I pick which path among all SP when there is a choice?

Dijkstra, where path length l replaced by tuple (l, m)
 $\uparrow \quad \uparrow$
 path length slope

$\rightarrow (l_1, m_1) \leq (l_2, m_2)$ by dictionary order

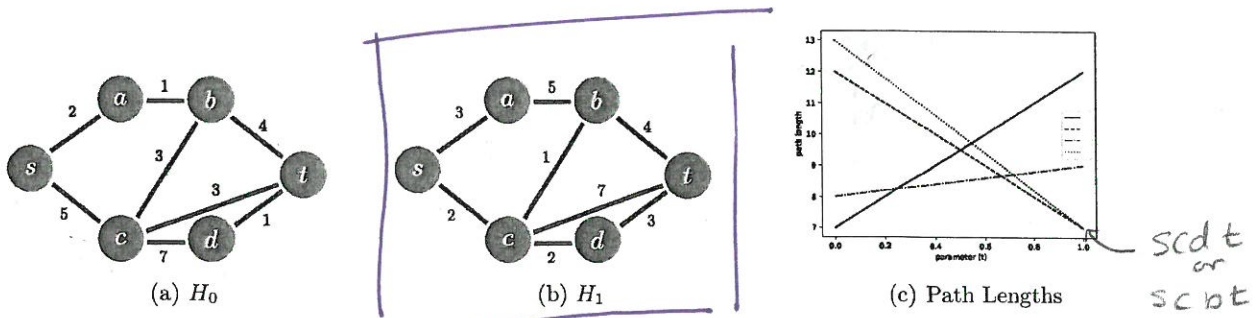


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